

PageRank and Random Walks

Ian Chen

University of Illinois Urbana-Champaign

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Nibbling towards a Partition

- Our goal is to find a set S , $2 \operatorname{vol}(S) \leq \operatorname{vol}(V)$, minimizing $\Phi(S) = \frac{|\partial S|}{\operatorname{vol}(S)}$.
- Spectral partitioning finds an $O(\sqrt{\Phi_{\text{OPT}}})$ approximation (Cheeger)
- Let $v \in C$ be a seed, $\alpha = \tilde{O}(\Phi(C))$, $\epsilon = \tilde{O}(\frac{1}{\operatorname{vol}(C)})$
- $\text{PR-NIBBLE}(v, \alpha, \epsilon)$ finds an $\tilde{O}(\sqrt{\alpha})$ -cut in time $\tilde{O}(1/\epsilon\alpha)$
 - ▶ Linear in the size of the output (for fixed conductance)
 - ▶ Sampling seeds, binary searching on ϵ and α , finds approximately optimal global cut

Personalized PageRank

- On an undirected graph, the walk matrix $W = AD^{-1}$
- Lazy walk matrix $\tilde{W} = \frac{1}{2}(I + AD^{-1})$ is aperiodic
- Stationary distribution is $\psi = \frac{\mathbf{d}}{\text{vol}(V)}$
- Teleportation α with starting $\mathbf{s}$, $\hat{W} = \alpha\mathbf{s} + (1 - \alpha)\tilde{W}$ is ergodic
- Let $\mathbf{s} = \mathcal{X}_v$; stationary distribution measures locality of u wrt v

$$\begin{aligned}\pi_{\alpha, \mathbf{s}} &= \alpha\mathbf{s} + (1 - \alpha)\tilde{W}\pi_{\alpha, \mathbf{s}} = \alpha(I - (1 - \alpha)\tilde{W})^{-1}\mathbf{s} \\ &= \alpha \sum_{t=0}^{\infty} (1 - \alpha)^t \tilde{W}^t \mathbf{s}\end{aligned}$$

Random Walks and Cuts

- Low conductance sets have small boundaries
- Low conductance sets have higher than expected PPR mass

Theorem

If $\pi(S) - \psi(S) > \alpha t + \sqrt{\min(\text{vol}(S), \text{vol}(\bar{S}))}(1 - \frac{\phi^2}{8})^t$, then $\exists S_j^\pi$ sweep s.t.

$$\Phi(S_j^\pi) \leq \phi$$

- Sample vertices v , estimate PPR, and compare conductance for sweep cuts.

Analysis

- Associate probability on the edges, $\mathbf{p}(uv) = \frac{\mathbf{p}(u)}{\deg(u)}$, e.g. $\psi(e) \equiv \frac{1}{2m}$
- Let $\mathbf{p}\{k\}$ be the mass on the k largest entries of $\mathbf{p}$, e.g. $\psi\{k\} = \frac{k}{2m}$
- Lovasz-Simonovitz curve extends $\mathbf{p}\{k\}$ to $[0, 2m]$ by linear interpolation
 - ▶ Concave, piecewise linear
 - ▶ $\mathbf{p}\{0\} = 0, \mathbf{p}\{2m\} = 1$
 - ▶ $\mathbf{p}\{x\} \geq \psi\{x\}$
- We are bounding the convergence of a random walk by a *curve*

Analysis (cont.)

- Denote $f_t(x)$ as $\alpha t + \sqrt{\min(x, 2m - x)}(1 - \frac{\phi^2}{8})^t$

Theorem

If $\nexists S_j$ s.t. $\Phi(S_j) < \phi$ and $\pi(S_j) - \psi(S_j) > f_t(\text{vol}(S_j))$, then $\pi\{x\} - \frac{x}{2m} \leq f_t(x)$.

- We prove this for $x \leq m$; other case is similar
- Base case: $t = 0$, follows by having slope bounded by 1

$$\begin{aligned}\pi\{x\} - \frac{x}{2m} &\leq \min(1, \min(x, 2m - x)) \\ &\leq \sqrt{\min(x, 2m - x)}\end{aligned}$$

Analysis (cont.)

- Study the distribution of mass after one step of $\tilde{W}$

$$\begin{aligned}(\tilde{W}\pi)(S) &= \frac{1}{2}p(E(S, V)) + \frac{1}{2}p(E(V, S)) \\ &= \frac{1}{2}p(E(S, S)) + \frac{1}{2}p(E(S, V) \cup E(V, S))\end{aligned}$$

- $|E(S)| = \text{vol}(V) - |\partial S| = \text{vol}(V)(1 - \phi)$
- $|E(S, V) \cup E(V, S)| = \text{vol}(V) + |\partial S| = \text{vol}(V)(1 + \phi)$
- We constructed π to be stationary wrt the PageRank walk:

$$\pi \{ \text{vol}(S) \} \leq \alpha + \frac{1}{2}(\pi \{ \text{vol}(S)(1 - \phi) \} + \pi \{ \text{vol}(S)(1 + \phi) \})$$

Analysis (cont.)

- Plugging in the inductive hypothesis, then

$$\begin{aligned}\pi \{ \text{vol}(S) \} &\leq \alpha + (t-1)\alpha + \frac{\sqrt{\text{vol}(S)}}{2}(\sqrt{1-\phi} + \sqrt{1+\phi})\left(1 - \frac{\phi^2}{8}\right)^{t-1} \\ &\leq \alpha t + \sqrt{\text{vol}(S)}\left(1 - \frac{\phi^2}{8}\right)^t\end{aligned}$$

Remarks

- Lovasz-Simonovitz curve used for mixing of $\tilde{W}$ (convex volume est.)
- We showed that $\pi \{ \text{vol}(S) \}$ is locally smoothed, by the conductance
- If $\mathcal{L} = D^{-1/2}LD^{-1/2} = \sum_{i=0}^{n-1} \lambda_i \mathbf{v}_i \mathbf{v}_i^T$,
 - ▶ The Green's function $\mathcal{G} = \sum_{i=1}^{n-1} \frac{1}{\lambda_i} \mathbf{v}_i \mathbf{v}_i^T$
 - ▶ Power iteration amplifies the signals with small eigenvalues
 - ▶ PPR normalizes $\beta I + \mathcal{L}$ (shift-inverse method) so that

$$\mathcal{G}_\beta = \sum_{i=1}^{n-1} \frac{1}{\lambda_i + \beta} \mathbf{v}_i \mathbf{v}_i^T$$

- The PPR vector we compute is given by (for $\beta = (2\alpha)/(1 - \alpha)$):

$$\frac{\pi_{\alpha, \mathbf{s}}}{\beta} = D^{-1/2} \mathcal{G}_\beta D^{1/2} \mathbf{s}$$